

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(SHORT SEMESTER EXAMINATION)

CLASS: BE
BRANCH: ALL

SEMESTER : SHORT
SESSION : SS/19

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

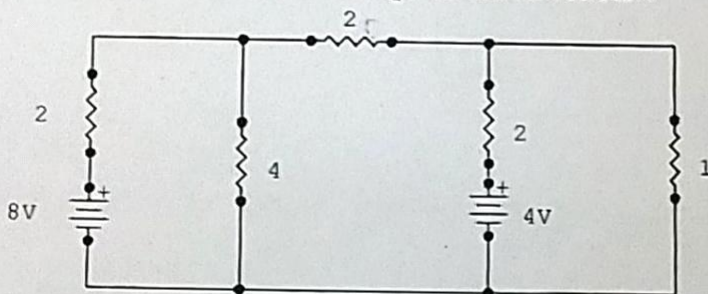
TIME: 03:00 Hours

FULL MARKS: 50

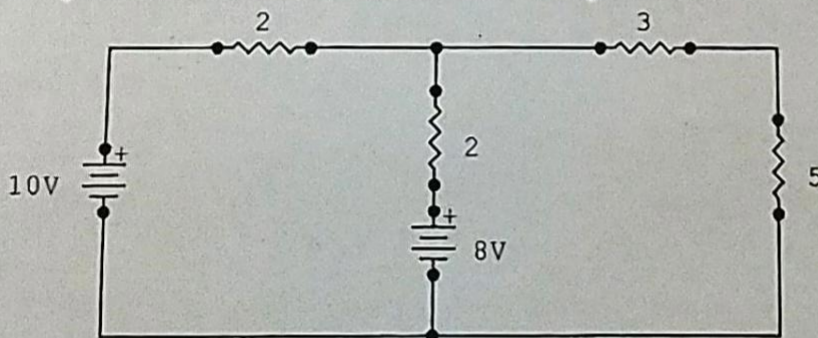
INSTRUCTIONS:

1. The question paper contains 7 questions each of 10 marks and total 70 marks.
2. Candidates may attempt any 5 questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

- Q.1(a) Distinguish between mesh and loop in electrical circuit. [2]
 (b) Write the expression for converting a delta network into star network and expression for converting a star network into delta network. [3]
 (c) Determine the current & voltage across 1Ω resistor. [5]



- Q.2(a) An alternating current is represented by $i = 10 \sin 942t$ Amp. Determine: (a) Frequency (b) Period [2]
 (b) Differentiate between series & parallel resonance. [3]
 (c) A 100Ω resistor shunted by a $0.4H$ inductor, is in series with capacitor C . A voltage of $250V$ at $50Hz$ is applied to the circuit. Find (A) The value of C to give unity power factor. (b) The total current. (c) The current in the inductive branch. [5]
- Q.3(a) Define the following: (i) Time period (ii) Frequency. [2]
 (b) Two voltages are given by: $V_1 = 50 \sin (377t - 30^\circ)$ & $V_2 = 50 \sin (377t + 45^\circ)$ act in series circuit. Determine: (a) Frequency of the total voltage (b) RMS value of the total. [3]
 (c) A voltage of $200V$ is applied to a series circuit consisting of a resistor, an inductor & a capacitor. The respective voltages across these components are $170V$, $150V$ & $100V$ and the current is $4A$. Find the power factor of the inductor of the circuit. [5]
- Q.4(a) State & explain Thevenin's Theorem. [2]
 (b) Derive relations between line and phase voltages, in star and delta connections. [3]
 (c) Using Norton's Theorem, find current through 5Ω resistor. [5]



3

- Q.5(a) Give the Line and Phase relation for Delta connection. [2]
 (b) A symmetrical three phase, $400V$ system supplies a balanced delta connected load. The current in each delta branch is $30A$ & its phase angle is 37° lagging. Find the line current & total power. [3]
 (c) A balanced delta connected load of $16 + j12\Omega$ per phase is connected to a three phase, $230V$ supply. Find the line current, power factor, power, Reactive VA & Total VA. [5]

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- Q.6(a) Define leakage factor in a magnetic circuit. [2]
(b) Give the analogy between electric and magnetic circuits. [3]
(c) A iron ring of circumference of 140 cm & cross section of 12 cm^2 is wound with 500 turns of wire. [5]
When I is 2 Amp, $\phi = 1.2 \text{ m wb}$. Find relative permeability of iron.
- Q.7(a) Write a short note on classification of INDICATING INSTRUMENTS. [2]
(b) Explain the role of damping torque in indicating instruments. [3]
(c) Describe constructional details of the Permanent Magnet Moving Coil (PMMC) instrument [5]

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INSTRUCTIONS:

1. The question paper contains 7 questions each of 12 marks and total 84 marks.
2. Candidates may attempt any 5 questions maximum of 60 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

Q.1(a) Distinguish between

(i) active and passive elements (ii) loop and mesh

[2]

Q.1(b) Determine the node voltages and the current through the resistors using mesh analysis for the network shown in Fig-1. [4]

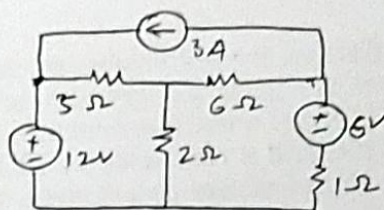


Fig-1

Q.1(c) Use mesh analysis to evaluate currents in the circuit shown in Fig. 2. [6]

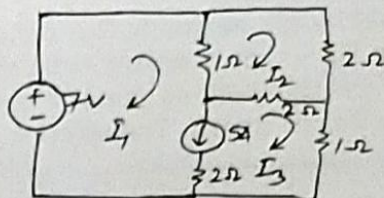


Fig-2

Q.2(a) The voltage and current in an AC circuit are given as $v = 100 \sin 314t$, $i = 10 \sin 314t$. Determine the average power and RMS power. [2]

Q.2(b) A non-alternating periodic waveform has been shown in Fig 3. Find its form factor. [4]

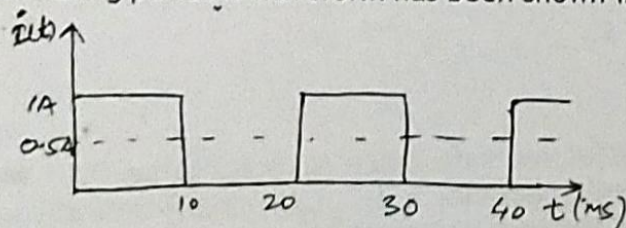


Fig-3

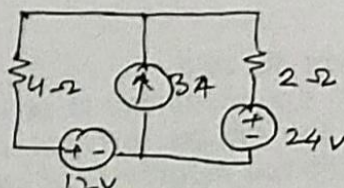


Fig-3

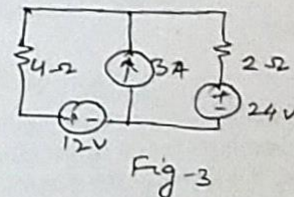
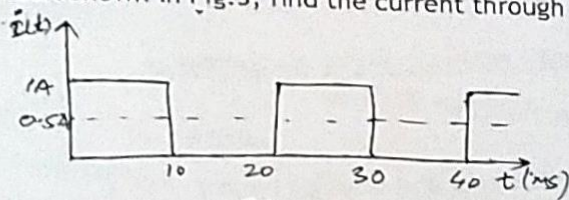
Q.2(c) In an inductive series circuit, a voltage of 10V at 25Hz produces 100mA while the same voltage at 75Hz produces 60mA. Find the values of circuit constants. At what frequency will the value of the impedance be twice that at 25Hz. [6]

Q.3(a) Explain the phasor method of solving AC parallel circuit (for two branches) with the help of an example. [2]

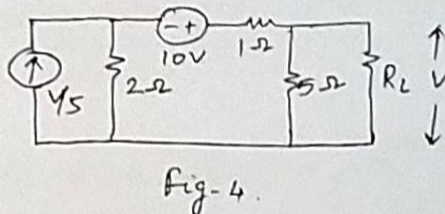
Q.3(b) Three impedances $(3+j5)$, $(4-j6)$ and $(8+j6)$ are connected in parallel. Calculate the current in each branch when the total current is 20A. [4]

Q.3(c) A RLC series circuit has $R=5\Omega$, $L=0.2H$ and $C=50\mu F$. The applied voltage is 200V. Find (i) resonant frequency (ii) Q factor (iii) bandwidth (iv) upper and lower half-power frequencies (v) current at resonance (vi) current at half-power points (vi) voltage across inductance at resonance. [6]

- Q.4(a) State and prove maximum power transfer theorem. [2]
 Q.4(b) For the circuit shown in Fig.3, find the current through $4\ \Omega$ resistor using superposition theorem. [4]

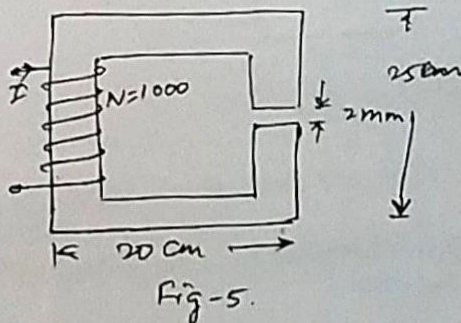


- Q.4(c) Using Thevenin's theorem determine the power loss in R_L , R_L is $10\ \Omega$ in the circuit given in Fig.4. [6]

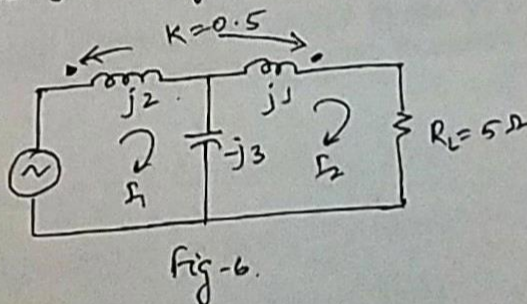


- Q.5(a) Specify the reason why three-phase systems are preferred over single-phase systems. [2]
 Q.5(b) Develop the expression for total power in a balanced three-phase star and delta load. [4]
 Q.5(c) A 3-phase star-connected alternator supplies a delta-connected load with a 415V, 50Hz supply. Each phase of the load has resistance of $15\ \Omega$ and a reactance of $25\ \Omega$. Calculate (a) the current in each phase of the load (b) the alternator phase voltage and current (c) the power factor of the load (d) the total power taken by the load. [6]
 Sketch the circuit diagram and show on it the phase and line voltages and currents.

- Q.6(a) Distinguish between electric and magnetic circuit. [2]
 Q.6(b) Define the following terms (i) permeability (ii) magnetic flux density (iii) MMF (iv) reluctance. [4]
 Q.6(c) A rectangular magnetic core shown in Fig 5. has a square cross-section of area 16cm^2 . An airgap of 2mm is cut across one of its limbs. Find the exciting current needed in the coil having 1000 turns wound on the core to create an airgap flux of 5mWb. The relative permeability of the core is 2000. [6]



- Q.7(a) State and explain dot rule. [2]
 Q.7(b) When two coils are connected in series, their effective inductance is found to be 14H. When the connections of one coil are reversed, the effective inductance is 8H. If the coefficient of coupling is 0.8, calculate the self inductance of each coil and the mutual inductance. [4]
 Q.7(c) Find the drop across R_L in the circuit given in Fig 6. [6]



CLASS : BE
BRANCH : ALL

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(NONCOLLEGIATE EXAMINATION)

SEMESTER : NC2017
SESSION : SP/2017

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME : 3 HOURS

INSTRUCTIONS :

FULL MARKS: 100

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Successful candidates are entitled to 'C' grade only.

Q.1(a) Define Real voltage & Real current source?

Q.1(b) Find an equivalent resistance between A & B in the network of fig-1. using star-delta transformation [4]
[6]

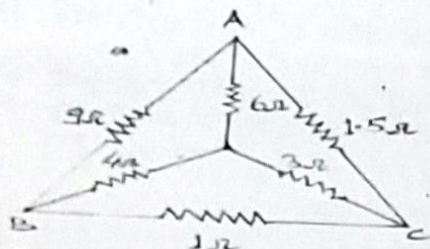


Fig-1.

Q.1(c) In the network of fig-2., find the node voltages V_1 , V_2 & V_3 using super-node analysis? [10]

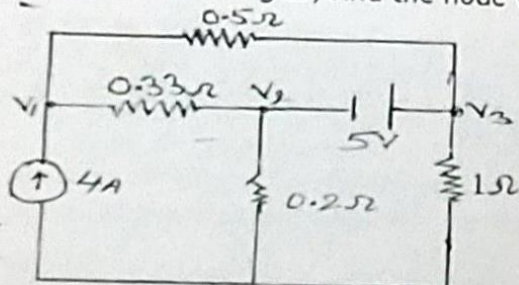


Fig-2.

- Q.2(a) Define Alternation, Amplitude, Form factor & Instantaneous value of A.C terminology? [4]
Q.2(b) Determine the Form factor of the saw-tooth wave shown in fig-3. ? [6]

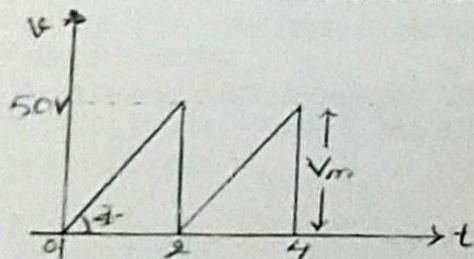


Fig-3.

- Q.2(c) A voltage $V(t) = 100\sin 314t$ is applied to a series circuit consisting of 10Ω resistance, $0.0318H$ Inductance and a capacitor of $63.6\mu F$. find (i) Expression for $i(t)$. (ii) Phase angle between voltage and current. (iii) Power factor. (iv) Active power consumed and (v) Peak value of pulsating power. [10]

Q.3(a) State that Acceptor & Rejector circuit ? [4]

Q.3(b) Find the value of R , L & C in a series RLC circuit that resonates at $1.5KHz$ and consumes $50W$ from a $50V$ a.c source operating at the resonant frequency. The bandwidth is $0.75KHz$. [6]

Q.3(c) Two circuits A & B are connected in parallel to a $115V$, $50Hz$ supply. The total current taken by the combination is $10A$ at unity power factor. Circuit A consists of a 10Ω resistor and $200\mu F$ capacitor connected in series. Circuit B consists of a resistor and an inductor in series, determine the current, power factor, impedance, resistance and reactance of branch B. [10]

- Q.4(a) State that Superposition's Theorem? [4]
 Q.4(b) State & prove the maximum power transfer theorem? [6]
 Q.4(c) Obtain the thevenin's equivalent network across terminals A & B in fig-4.? [10]

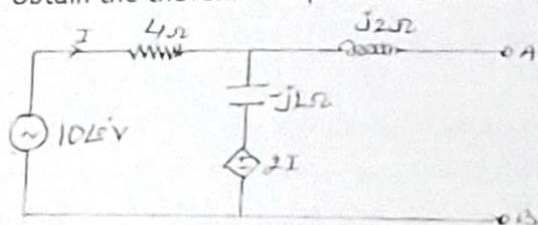


Fig-4.

- Q.5(a) What are the advantages of a 3- Φ system over 1- Φ system? [4]
 Q.5(b) Three identical coils connected in Delta to a 440V, 3- Φ supply take a total power of 50KW and a line current of 90A. Find the phase current, power factor and apparent power taken by the coils. [6]
 Q.5(c) A 3- Φ , 220V, 50Hz, 11.2KW induction motor has a full load efficiency of 88% and draws a line current of 38A under full load, when connected to a 3- Φ , 220V supply. Determine power factor at which the motor is operating. Find the reading on two wattmeters connected in the circuit to measure the input to the motor. [10]
 Q.6(a) Define Magnetic flux & magnetic flux density? [4]
 Q.6(b) Determine the m.m.f required to generate a total flux of 100 μ Wb in an air gap 0.2 c.m long, the cross sectional area of the air gap is 25 c.m². [6]
 Q.6(c) The magnetic circuit in fig-5. is built up of iron of square cross section 3 c.m side. Each airgap is 2mm wound with 1000 turns & exciting current is 1A. The relative permeability of part A and part B may be taken as 1000 & 1200 respectively. Find (i) Reluctance of part A. (ii) Reluctance of part B. (iii) Reluctance of two airgaps. (iv) Total Reluctance. (v) total m.m.f. [10]

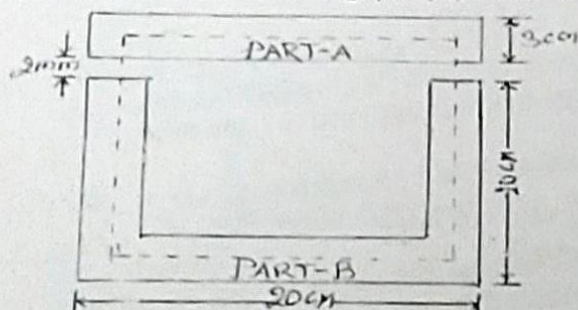


Fig-5.

- Q.7(a) State that self & mutually induced emf? [4]
 Q.7(b) Explain hysteresis loss & PMMC type instruments. [6]
 Q.7(c) In the network shown in fig-6. , find the voltages V_1 & V_2 [10]

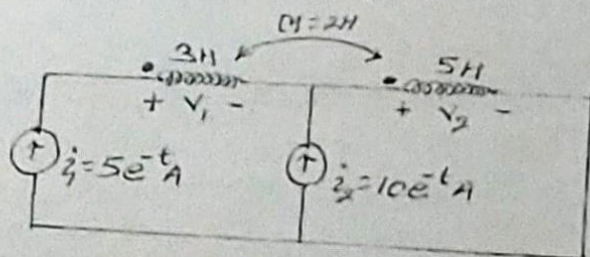


Fig-6.

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(NON-COLLEGIATE EXAMINATION)

CLASS: BE
BRANCH: EEE

SEMESTER : NC
SESSION : MO/17

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

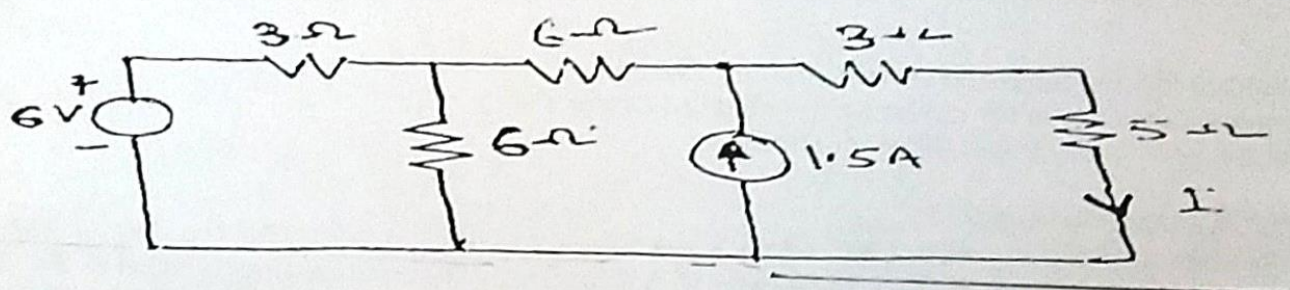
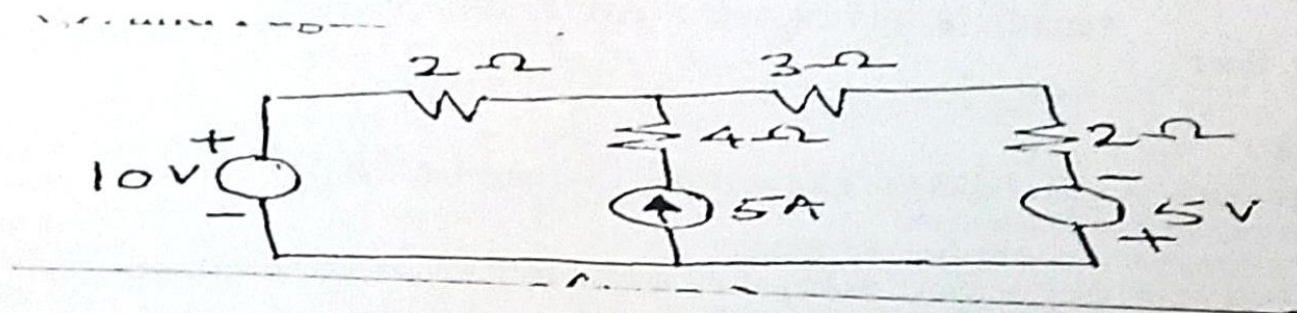
TIME: 3. hours

FULL MARKS: 100

INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
 2. Candidates may attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Before attempting the question paper, be sure that you have got the correct question paper.
 5. Successful candidates are entitled to 'C' grade only.
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- Q.1(a) What are ideal voltage and current sources? Explain. [4]
Q.1(b) Obtain an expression for converting star to delta connection. [6]
Q.1(c) Use nodal analysis to find different currents. [10]
- Q.2(a) What is a power triangle? [4]
Q.2(b) Two wires carrying currents i and i join a third wire and this then carries the sum of these two currents. $i_1 = 10 \sin \omega t$ & $i_2 = 15 \sin(\omega t + 45^\circ) + \cos 3 \omega t$. What will be the equation of the current i in the third wire? What will be reading of an a.c. ammeter when connected in each of the lines with the above currents flowing. [6]
Q.2(c) An inductive coil having a resistance of 15 ohm takes a current of 4A when connected to a 100-V, 60-Hz supply. If the coil is connected to a 100-V, 50-Hz supply, calculate (i) the current, (ii) the power (iii) the p.f. [10]
- Q.3(a) What are the different methods of solving a.c parallel circuits? [4]
Q.3(b) Plot the variation of current, X_L , X_C , R , Z with frequency in a series RLC circuit. [6]
Q.3(c) A coil having a resistance of 5 ohms and an inductance of 0.1H is connected in series with a condenser of capacitance 50 μF . A constant alternating voltage of 200V is applied to the circuit. At what value of the frequency will the current be maximum? Calculate this current and the voltage across coil and across capacitor for this frequency. Find also the voltage magnification at resonance. [10]
- Q.4(a) State and explain Thevenin's theorem. [4]
Q.4(b) State and explain Maximum power transfer Theorem for a.c. Circuits. [6]
Q.4(c) Find the current I in the figures shown below using Superposition theorem. [10]
- Q.5(a) What are the advantages of 3 phase system over single phase system? [4]
Q.5(b) Derive the relations between line and phase quantities of balanced 3 phase system in star connection. [6]
Q.5(c) A balanced delta connected load is supplied from a symmetrical three phase 400V system. The current in each phase is 30A and leads 30° behind the phase voltage find the active, reactive and apparent powers. [10]
- Q.6(a) Give examples showing diagrams of (a) series, (b) parallel and (c) series and parallel magnetic circuits. [4]
Q.6(b) What is the analogy between electric circuits and magnetic circuits? [6]
Q.6(c) A magnetic core, in the form of a closed ring, has a mean length of 20 cm and a cross-section of 1 sq. cm . The relative permeability of iron is 2400. What direct current will be needed in a coil of 2000 turns uniformly wound round the ring to create a flux of 0.2mWb in the iron? [10]
- Q.7(a) What are self and mutual inductances? [4]
Q.7(b) What are the methods of providing different types of torques? [6]
Q.7(c) Explain with the help of neat diagram the principle and working of a MI type of instrument. [10]



:::::22-05-2018 NC:::::

BIRLA INSTITUTE OF TECHNOLOGY, MESRA : RANCHI
(END SEMESTER EXAMINATION MO/12)

CLASS : B.E.
BRANCH : ALL

SEMESTER: I (DELAY BATCH)

SESSION : MO/12

Subject with Code : EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

FULL MARKS : 60

TIME : 3.00 HOURS

INSTRUCTIONS :

1. The question paper contains 7 questions each of 12 marks and total 84 marks.
2. Candidates may attempt any 5 questions maximum of 60 marks.
3. The missing data, if any may be assumed suitably.
4. Before attempting the question paper, be sure that you got a correct question paper.
5. Tables/Data hand book/Graph Paper etc. to be supplied to the candidates in the examination hall

- Q.1(a) Two coils are connected in series with same polarities and the combined inductance is found to be 0.567H. When the coils are connected in series with reverse polarities then the combined inductance is 0.267H. The self-inductance of one coil is 0.3H. Determine the mutual inductance and the coupling coefficient. [2]
- Q.1(b) The current waveform through a 0.5H inductor is shown in Fig.-1. Calculate the voltage across the inductor at 0ms, 8ms and 14ms. [4]
- Q.1(c) In Fig.-2 the galvanometer G has a conductance of 10S. Determine the current through the galvanometer using Thevenin's Theorem. [6]
- Q.2(a) Write the mesh equations for the circuit shown in Fig.-3. [2]
- Q.2(b) Determine the node voltage V_1 and V_2 in the circuit shown in Fig.-4. [4]
- Q.2(c) In the network shown in Fig.-5, find the current through R_L . Assume $V = 5V$. [6]
- Q.3(a) State and explain the Superposition theorem? [2]
- Q.3(b) Obtain the Norton equivalent for the given network in Fig.-6. [4]
- Q.3(c) In Fig.-7, obtain the value of Z_L such that maximum power transfer takes place from the source to Z_L . Assume $jX_{L1} = jX_{L2} = j8\Omega$; $jM = j5\Omega$; $V_1 = 10\angle 0^\circ$ volts and $V_2 = 10\angle 90^\circ$. [6]
- Q.4(a) Find the resonant frequency of the given network in Fig.-8. [2]
- Q.4(b) A RLC series circuit operates at a frequency of ω and the ratio of ω to the resonant frequency is α . Obtain an expression of admittance of the circuit in the terms of R , Q & α . [4]
- Q.4(c) Show what value of R_L in the circuit shown in Fig.-9 will make it resonant. [6]
- Q.5(a) Define the following terms: (i) Permeability (ii) Relative permeability (iii) Permeance (iv) Reluctance [2]
- Q.5(b) A magnetic circuit made of wrought iron is shown in Fig.-10. It has two air gaps, each 1mm wide. The cross-sectional area of the central limb is 6 cm^2 and each outer limb is 4 cm^2 . If the coil wound on the central limb has 500 turns. Calculate the exciting current required to produce a flux of 0.9 mWb in it. The magnetizing curve for wrought iron gives
- | | | |
|----------------------------|-------|------|
| $B\text{ (Wb/m}^2\text{)}$ | 1.125 | 1.5 |
| $H\text{ (At /m)}$ | 500 | 2000 |
- Q.5(c) The coil of a PMMC instrument has 50½ turns, on a former 18 mm wide, the effective length of the conductors being 25mm. It moves in a uniform field of flux density 0.5T. The control spring constant is 1.5×10^{-6} Nm/degree. Calculate the current required to produce a deflection of 100 degrees. [6]
- Q.6(a) Determine the average and rms values of a current given by $i = 10 + 5\cos(314t)$ A. [2]
- Q.6(b) An unbalanced delta connected load consists of $R_1 = 100\Omega$, $R_2 = 200\Omega$, and a coil of $L_3 = 0.3H$ with negligible resistance. This is fed through from 3 phase, 400V, and 50Hz source as shown in Fig.-11. Calculate (i) The total power in the system (ii) The total Volt Ampere Reactive. [4]
- Q.6(c) How much capacitive VAR must be provided by the capacitor bank to improve the power factor to 0.95 lagging in Fig.-12? [6]
- Q.7(a) A non-sinusoidal voltage is having a form factor of 1.2 and peak factor of 1.5. If the average value of the voltage is 10 V. Calculate (i) rms value (ii) maximum value. [2]
- Q.7(b) Two coils A and B are connected in series across a 240V, 50Hz supply. The resistance R_A of A is 5Ω and inductance L_B of B is 0.05H. The active and reactive powers are 3 KW and 2 KVAR, respectively. Find (i) the resistance R_B of coil B (ii) the inductance L_A of coil A. (iii) the voltage drop across each coil. [4]
- Q.7(c) Draw the phasor diagram for the circuit shown in Fig.-12 indicating terminal voltages V_1 and V_2 and the current. Find the value of (i) current (ii) V_1 and V_2 , and (iii) Power factor. [6]

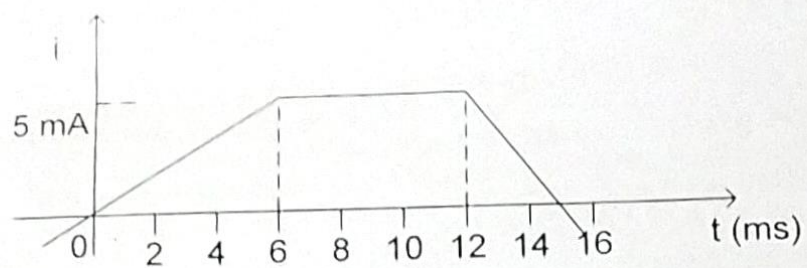


Fig-1

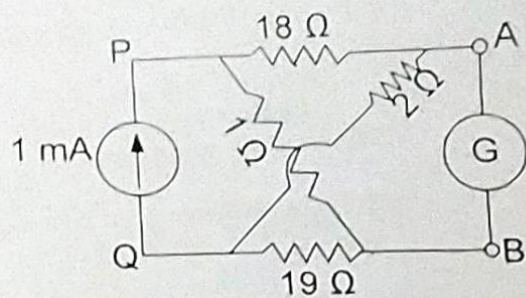


Fig-2

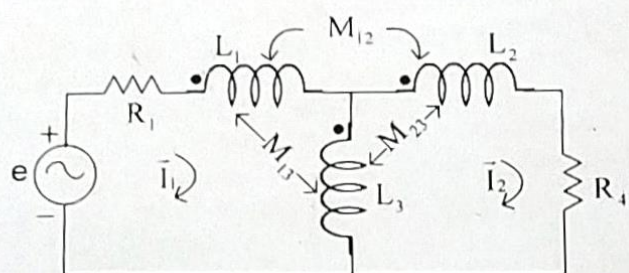


Fig-3

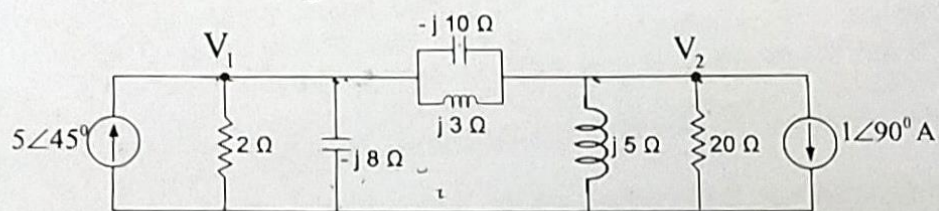


Fig-4

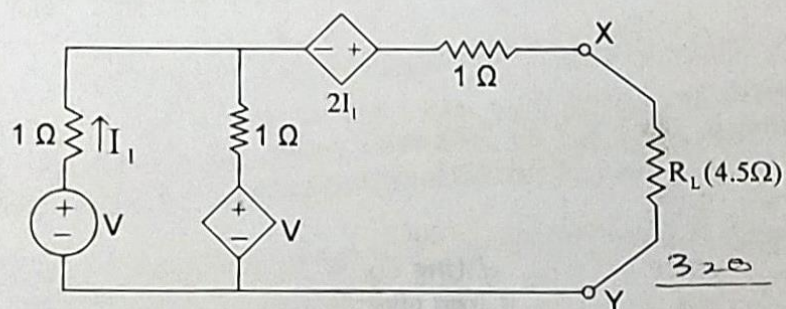


Fig-5

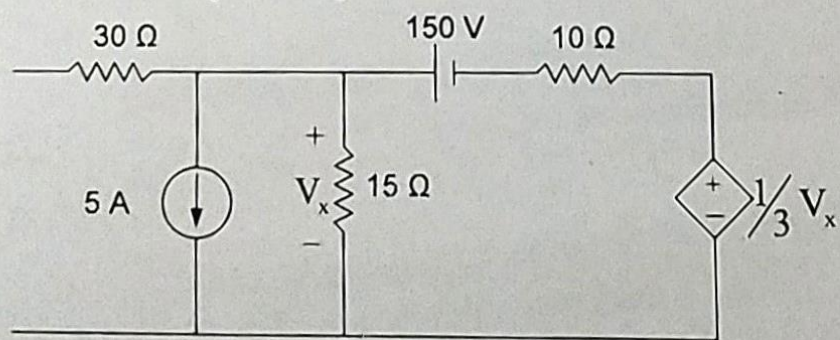


Fig-6

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS : B.E.
BRANCH : ALL

SEMESTER: II
SESSION : SP/2010

SUBJECT : BASIC ELECTRICAL ENGINEERING

TIME : 2 HOURS
INSTRUCTION :

FULL MARKS : 60

1. The questions paper contains 7 questions each of 12 marks and total 84 marks.
 2. Candidates may attempt any 5 questions maximum of 60 marks.
 3. The missing data, if any may be assumed suitably.
 4. Before attempting the question paper, be sure that you got a correct question paper.
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Q.1(a) Use mesh analysis to find the power delivered by the sources.
(b) Using Thevenin's theorem, find the current I_0 , flowing through the 1 ohm resistor. [6+6]

Q.2(a) Using Nodal analysis find the power dissipated in the 2 ohm resistor.
(b) Using Branch current method solve the following circuit and the power consumed in the network. [6+6]

Q.3(a) Using Superposition theorem find the voltage V_x in the circuit shown below.
(b) In the ring specimen find the current 'I' required to maintain a flux density of 1.2T in the air gap. [6+6]

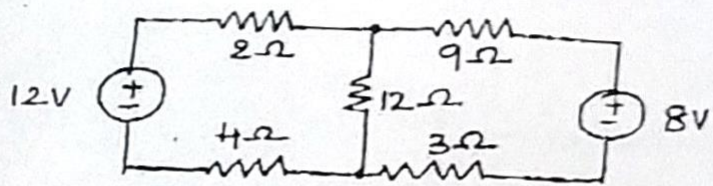
Q.4(a) In the following circuit prove that the power supplied by the sources is same as the power dissipated in the passive elements.
(b) Calculate $\bar{I}_1$ and $\bar{I}_2$ in the following circuit and draw a phasor diagram showing source voltage and all branch currents. [6+6]

Q.5(a) Determine the Thevenin Equivalent of the circuit as seen from the terminals a-b.
(b) Calculate v_o in the circuit using super position theorem. [6+6]

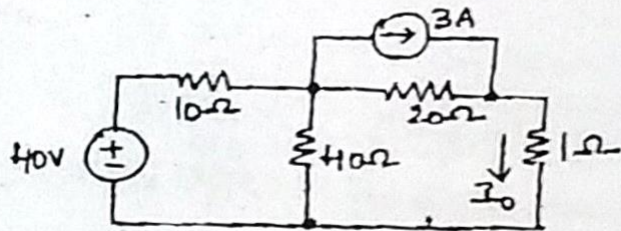
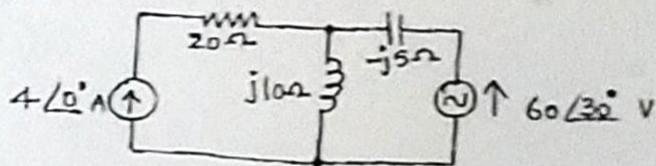
Q.6(a) Two coils are mutually coupled, with $L_1 = 25$ mH, $L_2 = 60$ mH, and $k = 0.5$. Calculate the maximum possible equivalent inductance if -
(a) the two coils are connected in series
(b) the coils are connected in parallel
(b) Find the Norton equivalent for the circuit, at terminals x-y. [6+6]

Q.7(a) Explain the principle of Operation of a PMMC Instrument. Derive the expression for deflecting Torque.
(b) A PMMC instrument has a resistance of 100 ohm and full scale deflection for a current of $400 \mu A$. Design a shunt arrangement with four resistance $R_1 = R_2 = R_3 = R_4 = 0.001$ ohms. [6+6]

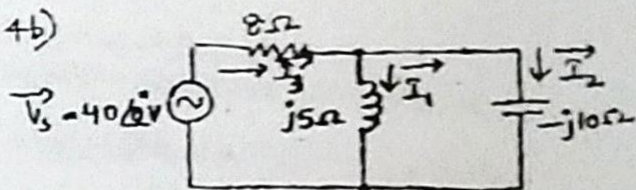
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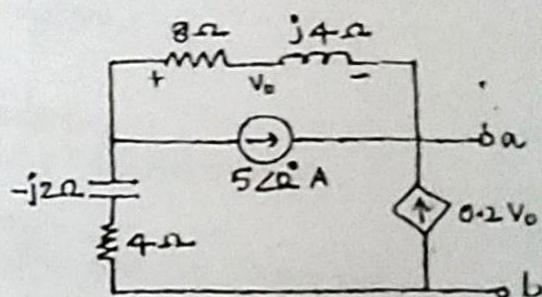
4a)



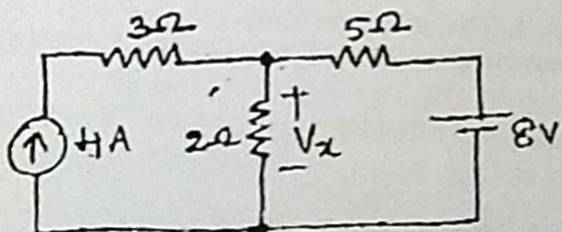
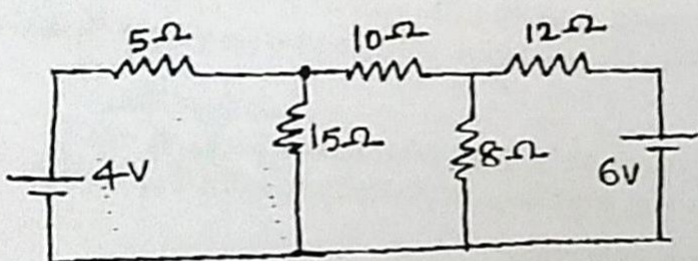
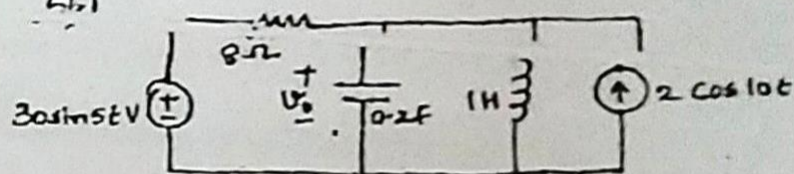
4b)



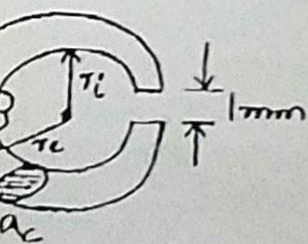
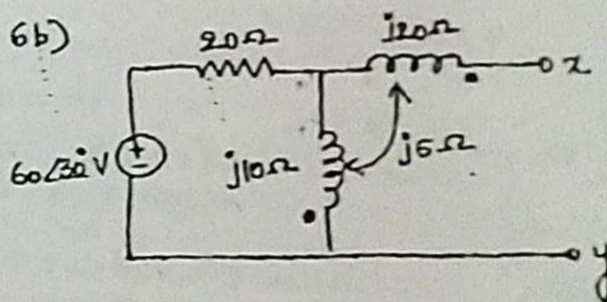
5a)



5b)



6b)



r_i = internal radius = 10mm

r_e = external radius = 12mm

a_c = area of cross-section of the core = 4mm²

N = No. of turns of the coil = 500

μ_r = Relative permeability = 4000

TIME : 3 HOURS

INSTRUCTIONS :

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Successful candidates are entitled to 'C' grade only.

- Q1. (a) Define (i) unidirectional element (ii) bidirectional element [4]
(b) Explain (i) supermesh analysis (ii) supernode analysis. [6]
(c) Use nodal method to find the current in various resistors of the circuit shown in Fig-1. [10]

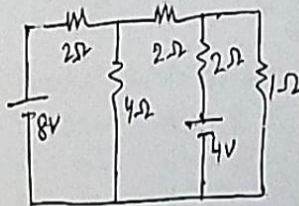


Fig-1.

- Q2. (a) Define form factor and peak factor of an alternating quantity. [4]
(b) Draw the following waves on a suitable scale: [6]
 $v = 200\sin(314t)$ and
 $i = 10\sin(314t - 60^\circ)$

Determine (i) RMS value of voltage and average value of current (ii) periodic time and frequency (iii) instantaneous value of voltage, when instantaneous value of current is -2.5A.

- (c) When a resistor and an inductor in series are connected to 240V, 50Hz AC mains, a current of 5A flows lagging 30° behind the supply voltage, while the voltage across the inductor is 200V. Determine the resistance of the resistor and the resistance and reactance of the inductor. Draw the phasor diagram of the circuit and impedance triangle. [10]

- Q3. (a) Explain the admittance method for solving AC parallel circuit with the help of an example. [4]
(b) Three impedances $(2+j5)$, $(3-j6)$ and $(8+j10)$ are connected in parallel. Calculate the current in each branch when the total current is 20A. [6]
(c) A RLC series circuit has $R=10\Omega$, $L=0.5H$ and $C=50\mu F$. The applied voltage is 200V. Find (i) resonant frequency (ii) Q factor (iii) bandwidth (iv) upper and lower half-power frequencies (v) current at resonance (vi) current at half-power points (vi) voltage across inductance at resonance. [10]

- Q4. (a) State Norton's theorem. [4]
(b) State Thevenin's theorem. Establish the similarity between Thevenin's and Norton's theorem. [6]
(c) Use Thevenin's theorem to determine the current through and the voltage across the 25Ω resistor given in Fig-2. [10]

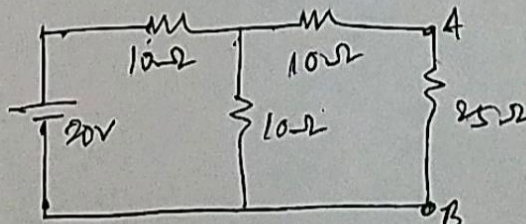


Fig-2

5. (a) Differentiate between balanced and unbalanced load for 3-phase system. [4]
 (b) Derive the relation between phase and line voltages and currents for a balanced three-phase star-connected system. [6]
 (c) A 3-phase delta connected load, each phase of which has an inductive reactance of 20Ω and a resistance of 25Ω , is fed from the secondary of a 3-phase star-connected transformer, which has a phase voltage of 240V. Draw the circuit diagram of the system and calculate (a) the current in each phase of the load (b) the p.d. across each phase of the load (c) the current in the transformer secondary winding (d) the total power taken from the supply and its power factor. [10]
6. (a) Explain fringing effect in magnetic circuits. [4]
 (b) Deduce the analogy between electric and magnetic circuits. [6]
 (c) Define the following terms: coercive force, Reluctance, remanence, residual flux density. [10]
7. (a) Distinguish between dynamically induced EMF and statically induced EMF. [4]
 (b) Explain the working principle of moving coil instrument with the help of neat sketch. [6]
 (c) Describe the following in case of measuring instruments [10]
 (i) Deflecting torque
 (ii) Controlling torque
 (iii) Damping torque

:::::: 29/08/2018 :::::

CLASS : BE
BRANCH : All
TIME : 1.30 hrs.
SUBJECT : Basic Electrical Engineering
INSTRUCTIONS:

SEMESTER : III
SESSION : SP 2008
FULL MARKS : 30

1. The total value of the question is 30 marks.
2. Candidate may attempt for all 30 marks.
3. In those case where the marks obtained exceed 25 marks, the excess marks will be ignored.

Q1. State and explain Superposition Theorem.

Q2. Find the equivalent resistance to the left of XY. fig1

Q3. Write down the mesh equations and find the different branch currents. fig2

Q4. Find the current I using Thevenin's theorem. Fig3

Q5. Find the current I using source conversion method. fig4

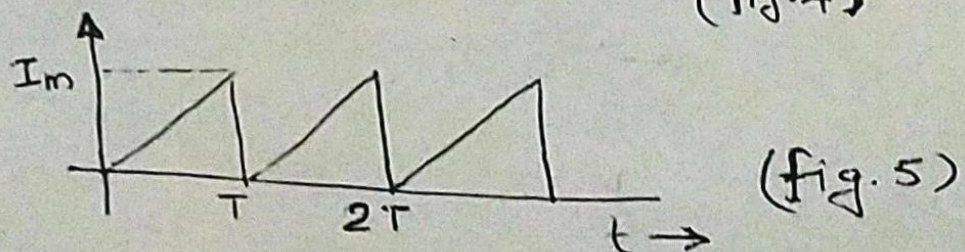
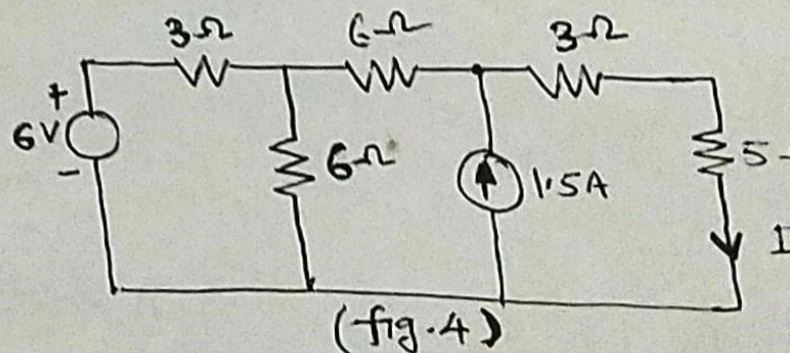
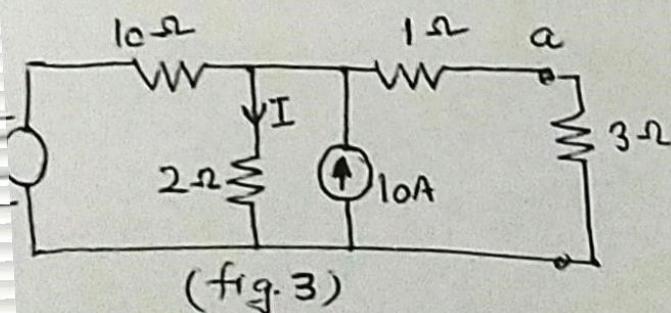
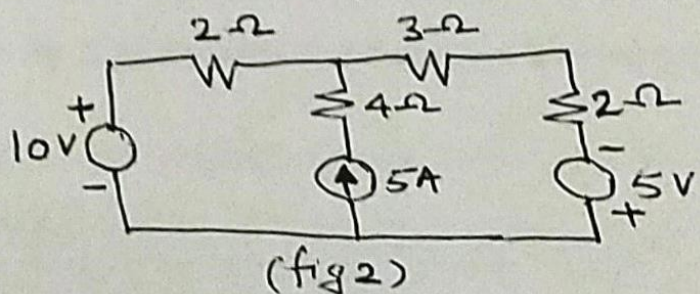
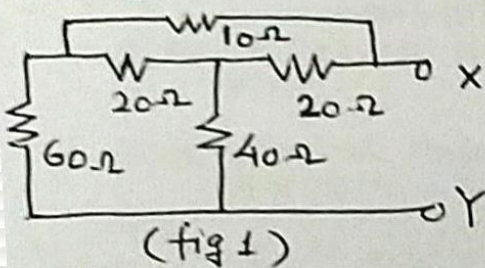
Q6. Find the rms and average value of given waveform. Also find the form factor. Fig 5

Q7. What is active power, reactive power and apparent power.

Q8. When a 400 V, 50Hz voltage is applied across a series combination of A resistance of 75 ohms and an inductor, the power dissipated is 1000 Watts. Find the impedance, reactance, inductance, power factor and the phase angle of the circuit.

Q9. Find the branch admittances and total admittance of a parallel circuit having impedances $8+j5$ and $6-j10$ and $5+j8$. Whether the circuit will behave as an R-L or R-C circuit.

Q10. Plot the variation of R, X_L, X_C, X, Z and I against frequency in R-L-C series circuit..



BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BTECH
BRANCH: CIVIL/CHEM. ENGG./CHEM. & POLY./BIOTECH/MECH/PROD/

SEMESTER: I/BL
SESSION : MO/2019

SUBJECT : EE101 BASIC ELECTRICAL ENGG.

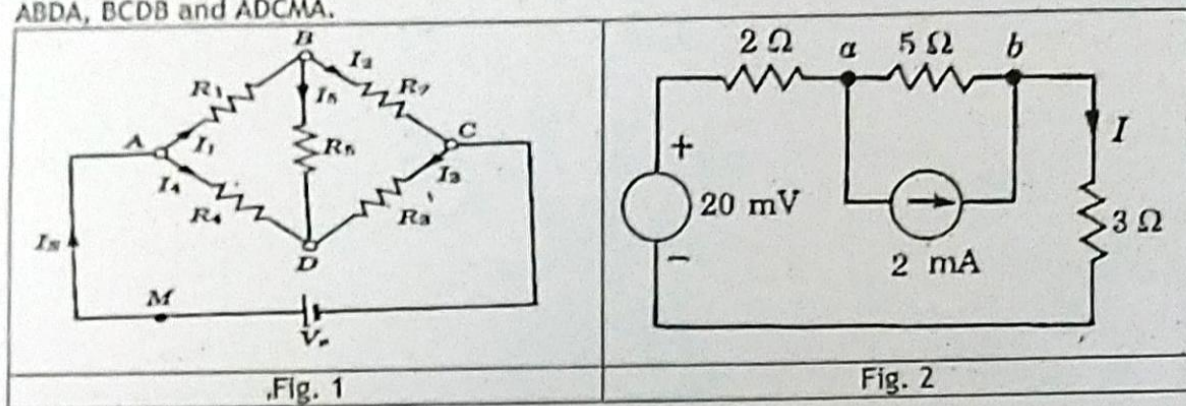
TIME: 2.00 HOURS

FULL MARKS: 25

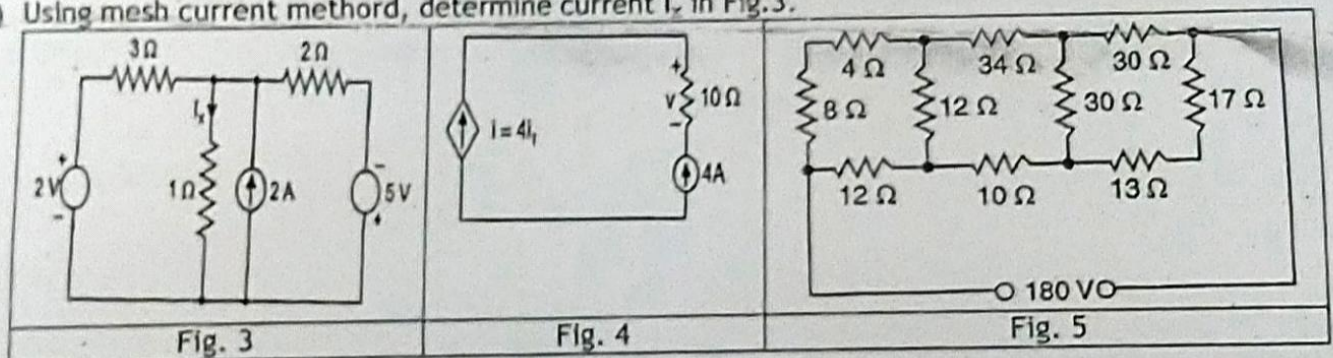
INSTRUCTIONS:

1. The total marks of the questions are 25.
2. Candidates may attempt for all 25 marks.
3. Before attempting the question paper, be sure that you have got the correct question paper.
4. The missing data, if any, may be assumed suitably.

- Q1 (a) State KVL and KCL and also give one example for each. [2]
Q1 (b) with the current as marked in Fig. 1, (i) write KCL at the four nodes; (ii) write KVL in meshes ABDA, BCDB and ADCMA. [3]



- Q2 (a) Use source transformation to find current I in Fig. 2. [2]
Q2 (b) Using mesh current method, determine current I_x in Fig. 3. [3]



- Q3 (a) Find the voltage v across the 10Ω resistor in Fig. 4, if the control current I_x in the dependent current source is (i) $2A$ and (ii) $-1A$. [2]
Q3 (b) Find the current in 10Ω resistor in Fig. 5 by star-delta transformation. Draw each conversion network. [3]
Q4 (a) Draw circuit diagram, phasor diagram and impedance triangle of (i) series RL circuit and (ii) parallel RC circuit. [2]
Q4 (b) A series circuit of a 300Ω non-inductive resistor, a $7.95 \mu F$ capacitor and a $2.06 H$ inductor of negligible resistance. If the supply voltage is $250 V$ at $50 Hz$, calculate: (i) the circuit current; (ii) the phase angle and (iii) the voltage drop across each element. [3]
Q5 (a) Draw variation of R , X , X_L , X_C and $-X_C$ with frequency for a series resonance circuit. [2]
Q5 (b) A $10 mH$ coil is connected in series with a loss-free capacitor to a variable frequency source which supplies a constant voltage of $10 V$. The circuit current has a maximum values of $0.1 A$ at a frequency of $80 kHz$. Calculate (i) the capacitance of the capacitor, (ii) the Q factor of the coil, and (iii) the half power frequency. [3]

: : : : : 16/10/2019 M : : : : :

CLASS: BE
BRANCH: ALL

SEMESTER : BACKLOG
SESSION : MO/19

SUBJECT: EE101 BASIC OF ELECTRICAL ENGINEERING

TIME: 3.00Hrs.

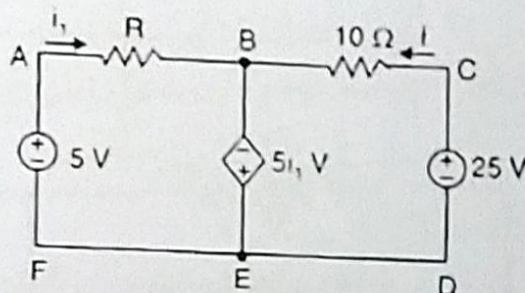
FULL MARKS: 50

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

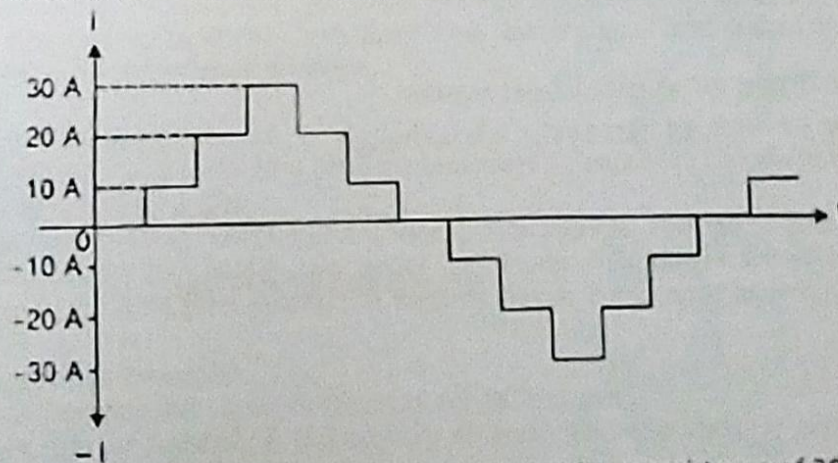
Q.1(a) Find the value of I in the circuit if $R = 10 \Omega$

[5]



Q.1(b) An iron ring has a mean diameter of 15 cm, a cross-section of 20 cm^2 and a radial gap of 0.5 mm cut in it. It is uniformly wound with 1500 turns of insulated wire and a magnetizing current of 1 A produces a flux of 1 mWb . Neglecting the effect of magnetic leakage and fringing. Calculate the reluctance of the magnetic circuit. [5]

Q.2(a) A current has the following steady values in amperes for equal interval of time changing instantaneously from one value to the next 0, 10, 20, 30, 20, 10, 0, -10, -20, -30, -20, -10, 0, etc. calculate (i) average value, (ii) r.m.s. value, (iii) form factor, and (iv) peak factor [5]

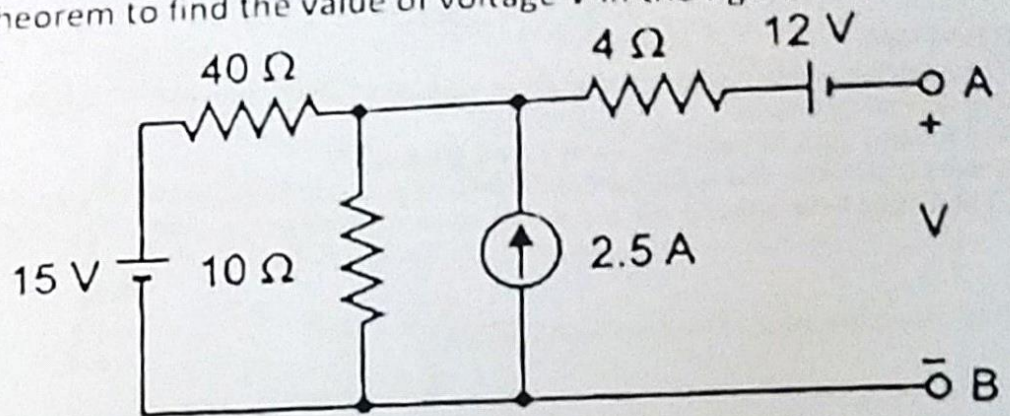


Q.2(b) A capacitor of $50 \mu\text{F}$ is connected in parallel with a coil that has a resistance of 20Ω and inductance of 0.05 H . If this parallel combination is connected across 200 V, 50 Hz supply, calculate (i) the line current, (ii) power factor and (iii) power consumed. [5]

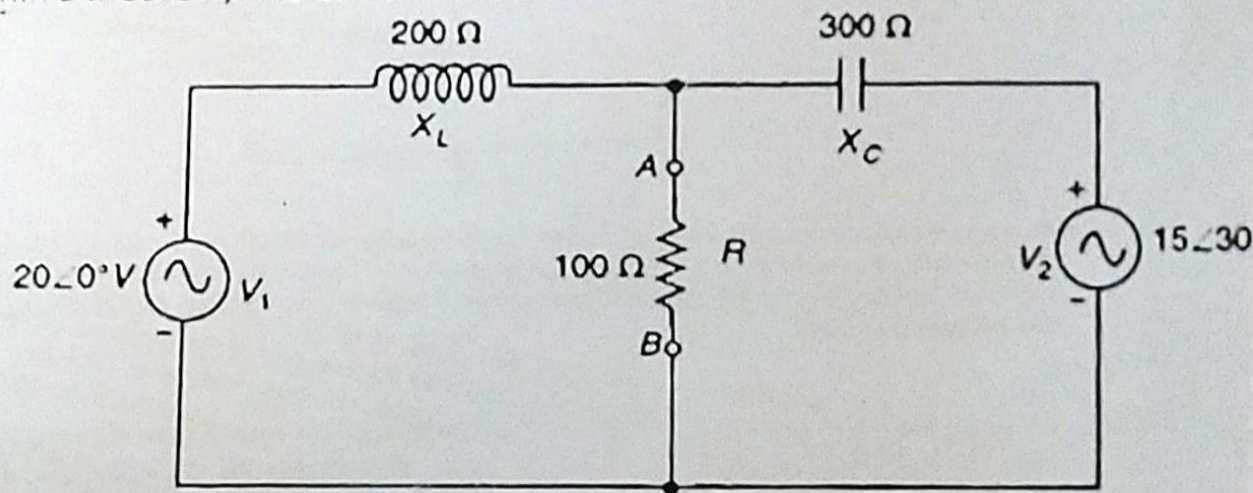
Q.3(a) Three 50Ω resistance are connected in star across 400 V, 3-phase supply. [5]
(i) Find phase current, line current and power taken from the main.
(ii) What would be the above values if one of the resistances were disconnected?

Q.3(b) In a 2-wattmeter method, power measured was 30 kW at 0.7 p.f. lagging. Find the reading of each wattmeter. [5]

2.4(a) Use superposition theorem to find the value of voltage V in the figure.



4(b) Using Thevenin's theorem, find current in the resistor R .



- Describe any two type of digital voltmeter.
- working principle of induction type wattmeter

:::::03/12/2019M:::::

CLASS: BE
BRANCH: ALL

SEMESTER : BACKLOG
SESSION : MO/19

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

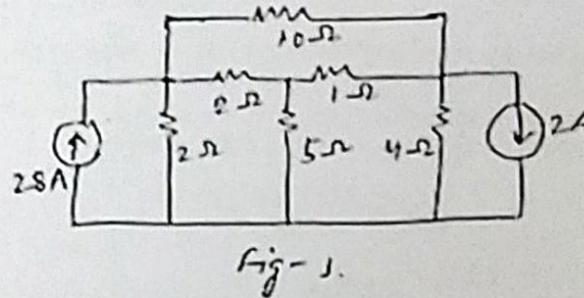
TIME: 3.00Hrs.

FULL MARKS: 100

INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt any 5 questions maximum of 100 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

- Q1. (a) Differentiate between (i) active and passive elements (ii) linear and nonlinear elements. [4]
Q1. (b) Explain Mesh current and Nodal voltage method. [6]
Q1. (c) Use nodal method to find the current in various resistors of the circuit shown in Fig-1. [10]



- Q2. (a) Define form factor and peak factor of an alternating quantity. [4]
Q2. (b) Draw the phasor diagram of the following voltages: [6]

$$v_1 = 100 \sin 100t$$

$$v_2 = 150 \sin(100t - \pi/6)$$

$$v_3 = 50 \sin(100t + \pi/4)$$

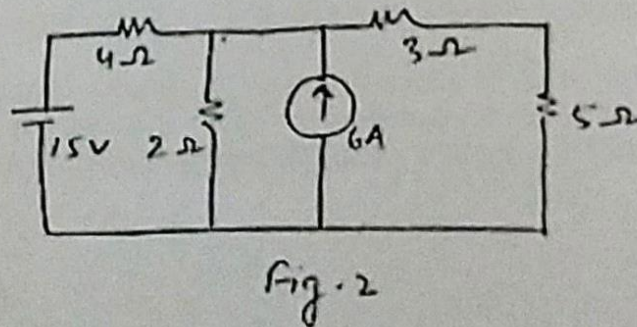
$$v_4 = 200 \cos 100t$$

Also find the resultant voltage and its RMS value.

- Q2. (c) When a resistor and an inductor in series are connected to 240V, 50Hz AC mains, a current of 2A flows lagging 30° behind the supply voltage, while the voltage across the inductor is 200V. Determine the resistance of the resistor and the resistance and reactance of the inductor. Draw the phasor diagram of the circuit and impedance triangle. [10]

- Q3. (a) Define power? Draw the power triangle. [4]
Q3. (b) Three impedances $(2+j4)$, $(3-j5)$ and $(8+j10)$ are connected in parallel. Calculate the current in each branch when the total current is 20A. [6]
Q3. (c) A RLC series circuit has $R=10\Omega$, $L=0.5H$ and $C=40\mu F$. The applied voltage is 250V. Find (i) resonant frequency (ii) Q factor (iii) bandwidth (iv) upper and lower half-power frequencies (v) current at resonance (vi) current at half-power points (vi) voltage across inductance at resonance. [10]

- Q4. (a) State and explain Norton's theorem. [4]
Q4. (b) State and prove maximum power transfer theorem for DC circuits. [6]
Q4. (c) Use Thevenin's theorem to determine the current through and the voltage across the 5Ω resistor given in Fig-2. [10]



- Q5. (a) Define (i) phase sequence (ii) balanced load for 3-phase system. [4]
 Q5. (b) Derive the relation between phase and line voltages and currents for a balanced three-phase delta-connected system. Draw the phasor diagram also. [6]
 Q5. (c) A 3-phase star-connected load, each phase of which has an inductive reactance of 10Ω and a resistance of 30Ω , is fed from the secondary of a 3-phase star-connected transformer, which has a phase voltage of 240V. Draw the circuit diagram of the system and calculate (a) the current in each phase of the load (b) the p.d. across each phase of the load (c) the current in the transformer secondary winding (d) the total power taken from the supply and its power factor. [10]
 Q6. (a) Explain (i) fringing effect (ii) leakage factor. [4]
 Q6. (b) Establish the analogy between electric and magnetic circuits. [6]
 Q6. (c) Explain Hysteresis and define the following terms: coercive force, Reluctance, remanence, residual flux density. [10]
 Q7. (a) Distinguish between dynamically induced EMF and statically induced EMF. [4]
 Q7. (b) Explain the working principle of moving iron instrument with the help of neat sketch. [6]
 Q7. (c) Define self inductance and mutual inductance. Obtain the expression for the mutual inductance of two coaxial solenoids of wires closely wound one upon the other. [10]

:::::10/12/2019M:::::

CLASS: BE
BRANCH: ALL

SEMESTER : 55/2019
SESSION : 2018-19

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

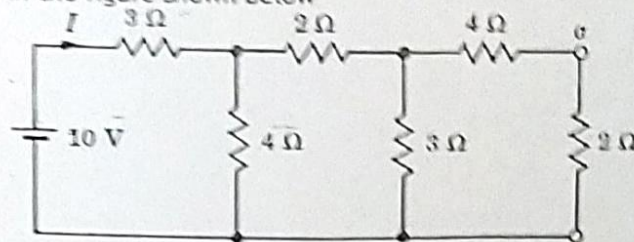
TIME: 3:00 HOURS

FULL MARKS: 100

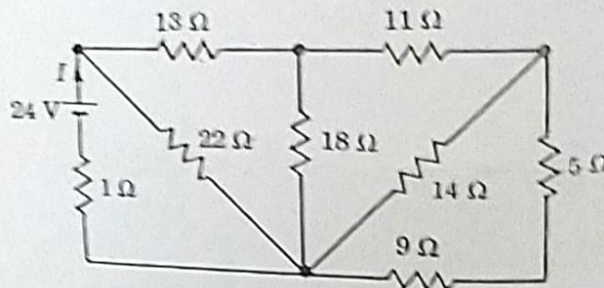
INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt any 5 questions maximum of 100 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

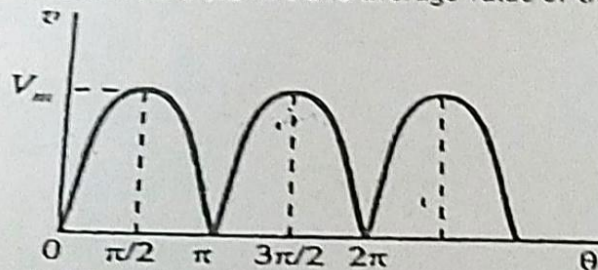
- Q.1(a) Define Kirchoff's Current Law (KCL) and Kirchoff's Voltage Law [4]
Q.1(b) Determine the current I in the figure shown below [6]



- Q.1(c) In the network shown below find (a) the current I (b) Voltage across 5Ω resistor [10]



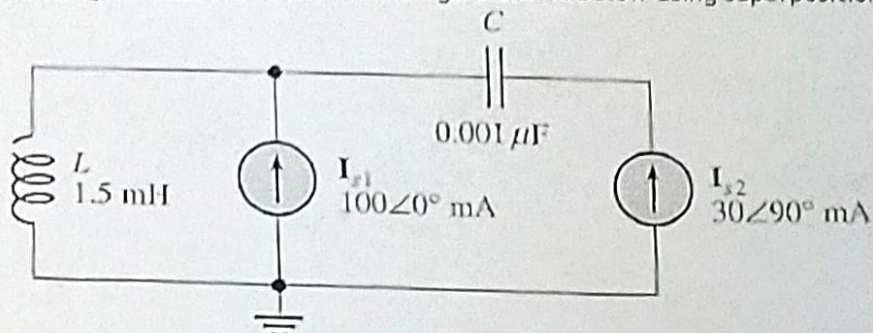
- Q.2(a) An alternating current is represented by $i = 10\sin 942t$ amperes. Determine (a) frequency (b) the period [4]
Q.2(b) A full rectified wave is shown below. Find the rms and average value of the voltage waveform [6]



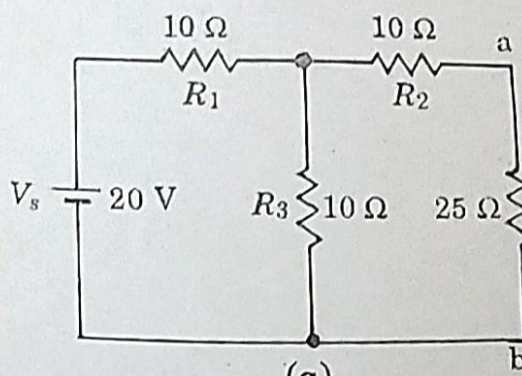
- Q.2(c) A resistance of 10Ω is connected in series with an inductance of $0.15H$ and a capacitance of $130\mu F$ to a $100V$, $50Hz$ a.c. supply. Calculate the (a) impedance (b) Current (c) phase angle (d) Power Factor. [10]
Q.3(a) What is Admittance, Conductance and Susceptance? [4]
Q.3(b) A 60Ω resistor connected in parallel with an inductive reactance of 80Ω to a $240V$, $50Hz$ supply. Calculate (a) the current through the resistor and inductive reactance (b) the supply current (c) circuit phase angle. [6]
Q.3(c) A circuit consists of a non-inductive resistor of 10Ω , an inductor of $0.1H$ and capacitor of $8\mu F$ in series. Calculate (a) Resonant frequency (b) current at resonant frequency (c) the voltage across each component when a voltage of $25V$ at resonant frequency is applied to the whole circuit? [10]

Q.4(a) State Maximum power transfer theorem? [4]

Q.4(b) Find the current through inductor $L=1.5\text{mH}$ in the figure shown below using superposition theorem [6]



Q.4(c) Use the venin's theorem to determine the current through and voltage across 25Ω resistor as shown in the figure below [10]



Q.5(a) Write the advantages of three phase circuits over single phase circuits [4]

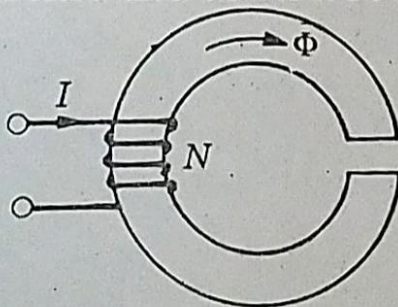
Q.5(b) Two wattmeters connected to measure the total power in a 3-phase load read 5kW and 2kW . Calculate the total power and the power factor? [6]

Q.5(c) A three phase load consists of three similar inductive coils, each of resistance 50Ω and inductance 0.3H are star-connected. The supply is 415V , 50Hz . Calculate (a) line current (b) the power factor [10]

Q.6(a) What is reluctance? Write the relation of MMF in terms of flux and reluctance? [4]

Q.6(b) Distinguish between magnetic and electric circuits? [6]

Q.6(c) A magnetic core in the form of a closed ring, has a mean length of 20cm and a cross-section of 1cm^2 . The relative permeability of iron is 2400 . What direct current will be needed in a coil of 2000 turns uniformly wound round the ring to create a flux of 0.2mWb in the iron? [10]



Q.7(a) Explain what is meant by self-Inductance and mutual inductance? [4]

Q.7(b) Describe the following [6]

(a) Deflecting torque

(b) Controlling torque

(c) Damping torque

Q.7(c) Discuss the advantages, limitations and applications of Moving Iron (MI) instruments [10]